2021 Ph H2 Q1

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs — Projectiles

A skier launches from a ramp at 16.0 m s^{-1} and 42.0°. (a) Find the horizontal and vertical components of the launch velocity. (b) Calculate the time to reach maximum height. (c) The skier then takes a further 1.40 s to land: calculate the horizontal distance travelled from the ramp to landing. (d) Compare kinetic energy just before landing with at take-off and justify.

Worked solution

(a) Components

$u = 16.0$ m s^{-1}, $\theta = 42.0°$.

Horizontal: $u_x = u \cos\theta = 16.0 \cos 42.0° = 11.89$ m s^{-1} (≈ 11.9).

Vertical: $u_y = u \sin\theta = 16.0 \sin 42.0° = 10.71$ m s^{-1} (≈ 10.7).

(b) Time to maximum height

At the top, $v_y = 0$, so $v_y = u_y - g\,t \Rightarrow t = u_y/g$.
$t = 10.71 / 9.8 = 1.09$ s.

Answer: 1.1 s

(c) Horizontal distance to landing

Total time = time up + time down = t_{up} + 1.40 s = 1.09 + 1.40 = 2.49 s.

Horizontal motion at constant u_x, so range s = $u_x \times t_{total}$.
s = 11.89 × 2.49 = 29.64 m.

Answer: 29.8 m (≈ 30 m)

(d) Kinetic energy comparison

Greater just before landing.

Justification: as the skier descends from the ramp to the ground, gravitational potential energy is converted to kinetic energy, so the speed — and hence KE — is greater at landing (ignoring air resistance).

Final answers

(a)(i) $u_x \approx 11.9$ m s^{-1}

(a)(ii) $u_y \approx 10.7$ m s^{-1}

(b) $t_{up} \approx 1.1$ s

(c) Horizontal distance ≈ 29.8 m (≈ 30 m)

(d) KE just before landing is greater (PE → KE).

Revision tips

- Resolve launch speed: $u_x = u \cos\theta$, $u_y = u \sin\theta$.
- At maximum height, $v_y = 0$, so $t_{up} = u_y/g$.
- Horizontal motion has constant speed if air resistance is neglected.
- Range = $u_x \times$ total time of flight.
- Energy: as height decreases, PE converts to KE (if no air resistance).